

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title: Big Basket E-commerce Database Management System

1. Database Design

1.1 Seller Table

Attribute	Key Type
Seller ID	Primary Key
Seller_Name	Not Null
Contact_Number	-
Email	-
Address	-

1.2 Inventory Table

Attribute	Key Type
Inventory_ID	Primary Key
Product_ID	Foreign Key
Seller_ID	Foreign Key
Stock_Quantity	Default
Stock_Status	Default
Last_Updated	Not Null

2. Primary and Foreign Key Relationship

Seller → Inventory

- Seller.SellerID → Primary Key
- Inventory.Inventory_ID → Primary Key
- Inventory.SellerID → Foreign Key referencing Seller.SellerID
- Inventory.Product_ID → Foreign Key referencing Product.Product_ID
- **Relationship:** One-to-Many (1:M) — one seller can have multiple inventory records.

3. Table Creation

3.1 Create Seller Table

```
CREATE TABLE Seller (  
    SellerID NUMBER PRIMARY KEY,  
    Seller_Name VARCHAR2(100) NOT NULL,  
    Contact_Number VARCHAR2(15),  
    Email VARCHAR2(100),  
    Address VARCHAR2(200)  
);
```

The Seller table is created with Seller ID as the primary key. It stores seller details such as seller name, contact number, email, and address.

3.2 Create Inventory Table

```
CREATE TABLE Inventory (  
    Inventory_ID NUMBER PRIMARY KEY,  
    Seller_ID NUMBER NOT NULL,  
    Product_ID NUMBER NOT NULL,  
    Stock_Quantity NUMBER DEFAULT 0,  
    Stock_Status VARCHAR2(20) DEFAULT 'In Stock',  
    Last_Updated DATE NOT NULL,  
    CONSTRAINT fk_seller FOREIGN KEY (Seller_ID) REFERENCES Seller(SellerID),  
    CONSTRAINT fk_product FOREIGN KEY (Product_ID) REFERENCES Product(Product_ID)  
);
```

The Inventory table is created with Inventory_ID as the primary key. SellerID and Product_ID are foreign keys that establish relationships with the Seller and Product tables.

4. Data Insertion

4.1 Seller Records

Five seller records are inserted into the Seller table, followed by a COMMIT to save the transaction

```

INSERT INTO Seller
VALUES (1, 'Fresh Mart', '9876543210', 'freshmart@gmail.com', 'Chennai');
1 row created.

INSERT INTO Seller
VALUES (2, 'Daily Needs', '9876543211', 'dailyneeds@gmail.com', 'Tambaram');
1 row created.

INSERT INTO Seller
VALUES (3, 'Green Basket', '9876543212', 'greenbasket@gmail.com', 'Velachery');
1 row created.

INSERT INTO Seller
VALUES (4, 'Smart Grocery', '9876543213', 'smartgrocery@gmail.com', 'Chromepet');
1 row created.

INSERT INTO Seller
VALUES (5, 'Nature Foods', '9876543214', 'naturefoods@gmail.com', 'Guindy');
1 row created.

```

4.2 Inventory Records

Inventory records are inserted with seller IDs, product IDs, stock quantities, stock status, and last updated dates.

```

INSERT INTO Inventory
VALUES (101, 1, 101, 50, 'In Stock',
|      | TO_DATE('16-09-2026', 'DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (102, 1, 102, 0, 'Out of Stock',
|      | TO_DATE('16-09-2026', 'DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (103, 2, 103, 30, 'In Stock',
|      | TO_DATE('16-09-2026', 'DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (104, 2, 104, 15, 'In Stock',
|      | TO_DATE('16-09-2026', 'DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (105, 3, 105, 0, 'Out of Stock',
|      | TO_DATE('16-09-2026', 'DD-MM-YYYY'));
1 row created.

```

```
INSERT INTO Inventory
VALUES (106, 3, 106, 25, 'In Stock',
|      | TO_DATE('16-09-2026', 'DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (107, 4, 107, 40, 'In Stock',
|      | TO_DATE('16-09-2026', 'DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (108, 4, 108, 0, 'Out of Stock',
|      | TO_DATE('16-09-2026', 'DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (109, 5, 109, 20, 'In Stock',
|      | TO_DATE('16-09-2026', 'DD-MM-YYYY'));
1 row created.

INSERT INTO Inventory
VALUES (110, 5, 110, 0, 'Out of Stock',
|      | TO_DATE('16-09-2026', 'DD-MM-YYYY'));
1 row created.
```

5. Data Retrieval

5.1 Display Seller Records

All seller records are retrieved to verify successful insertion.

```
Select * from Seller;
```

```
SELLERID
```

```
SELLER_NAME
```

```
CONTACT_NUMBER
```

```
EMAIL
```

```
ADDRESS
```

```
1  
Fresh Mart  
9876543210
```

```
SELLERID
```

```
SELLER_NAME
```

```
CONTACT_NUMBER
```

```
EMAIL
```

```
ADDRESS
```

```
freshmart@gmail.com  
Chennai
```

```
SELLERID
```

```
SELLER_NAME
```

```
CONTACT_NUMBER
```

```
EMAIL
```

```
ADDRESS
```

```
2  
Daily Needs  
9876543211
```

```
SELLERID
```

```
SELLER_NAME
```

```
CONTACT_NUMBER
```

```
EMAIL
```

```
ADDRESS
```

```
dailyneeds@gmail.com  
Tambaram
```

SELLERID

SELLER_NAME

CONTACT_NUMBER

EMAIL

ADDRESS

3
Green Basket
9876543212
SELLERID

SELLER_NAME

CONTACT_NUMBER

EMAIL

ADDRESS

greenbasket@gmail.com
Velachery

SELLERID

SELLER_NAME

CONTACT_NUMBER

EMAIL

ADDRESS

4
Smart Grocery
9876543213
SELLERID

SELLER_NAME

CONTACT_NUMBER

EMAIL

ADDRESS

smartgrocery@gmail.com
Chromepet

```

SELLERID
-----
SELLER_NAME
-----
CONTACT_NUMBER
-----
EMAIL
-----
ADDRESS
-----
| | | | 5
Nature Foods
9876543214
SELLERID
-----
SELLER_NAME
-----
CONTACT_NUMBER
-----
EMAIL
-----
ADDRESS
-----
naturefoods@gmail.com
Guindy
5 rows selected.

```

5.2 Display Inventory Records

All inventory records are retrieved to verify the stock information.

```

Select * from Inventory;

```

INVENTORY_ID	SELLER_ID	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
101	1	101	50	In Stock	16-SEP-26
102	1	102	0	Out of Stock	16-SEP-26
103	2	103	30	In Stock	16-SEP-26
104	2	104	15	In Stock	16-SEP-26
105	3	105	0	Out of Stock	16-SEP-26
106	3	106	25	In Stock	16-SEP-26
107	4	107	40	In Stock	16-SEP-26
108	4	108	0	Out of Stock	16-SEP-26
109	5	109	20	In Stock	16-SEP-26
110	5	110	0	Out of Stock	16-SEP-26

```

10 rows selected.

```

6. Seller Product Information

A JOIN query is used to display the seller name, product ID, stock quantity, stock status, and last updated date.

```
Select
s.SellerID,
p.Product_name,
i.Stock_Quantity,
i.Stock_Status
From Seller s
join Inventory i
ON s.SellerID=i.Seller_ID
join Product p
ON i.Product_ID=p.Product_ID
Order by s.Seller_Name;
```

SELLERID	PRODUCT_NAME	STOCK_QUANTITY

STOCK_STATUS		

	2 Tablet	30
In Stock		
	2 Keyboard	15
In Stock		
	1 Laptop	50
In Stock		
SELLERID	PRODUCT_NAME	STOCK_QUANTITY

STOCK_STATUS		

	1 Smartphone	0
Out of Stock		
	3 Mouse	0
Out of Stock		
	3 Monitor	25
In Stock		

SELLERID	PRODUCT_NAME	STOCK_QUANTITY

STOCK_STATUS		

5	Webcam	20
In Stock		
5	Headphones	0
Out of Stock		
4	Printer	40
In Stock		
SELLERID	PRODUCT_NAME	STOCK_QUANTITY

STOCK_STATUS		

4	Speaker	0
Out of Stock		
10 rows selected.		

7. Product Availability

7.1 Available Products

Products with stock available are displayed using the Stock_Status field.

```
Select
p.Product_ID,
p.Product_name,
i.Stock_Quantity,
i.Stock_Status
From Product p
join Inventory i
ON p.Product_ID=i.Product_ID
where i.Stock_Status='In Stock';
```

PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY

STOCK_STATUS		

101	Laptop	50
In Stock		
103	Tablet	30
In Stock		
104	Keyboard	15
In Stock		
PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY

STOCK_STATUS		

106	Monitor	25
In Stock		
107	Printer	40
In Stock		
109	Webcam	20
In Stock		
6 rows selected.		

7.2 Unavailable Products

Products with zero stock or Out of Stock status are displayed separately.

```
Select
p.Product_ID,
p.Product_name,
i.Stock_Quantity,
i.Stock_Status
From Product p
join Inventory i
ON p.Product_ID=i.Product_ID
where i.Stock_Status='Out of Stock';
```

PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY

STOCK_STATUS		

102	Smartphone	0
Out of Stock		
105	Mouse	0
Out of Stock		
108	Speaker	0
Out of Stock		
PRODUCT_ID	PRODUCT_NAME	STOCK_QUANTITY

STOCK_STATUS		

110	Headphones	0
Out of Stock		
4 row selected.		

8. Inventory Update

Demonstrate updating stock information

```
Select
p.Product_ID,
p.Product_name,
i.Stock_Quantity,
i.Stock_Status
From Product p
join Inventory i
ON p.Product_ID=i.Product_ID
where i.Stock_Status='Out of Stock';
```

```
UPDATE Inventory
SET Stock_Quantity = 30,
    Stock_Status = 'In Stock',
    Last_Updated = TO_DATE('17-09-2026', 'DD-MM-YYYY')
WHERE Inventory_ID = 110;
1 row updated.
```

```
UPDATE Inventory
SET Stock_Quantity = 10,
    Stock_Status = 'In Stock',
    Last_Updated = TO_DATE('17-09-2026', 'DD-MM-YYYY')
WHERE Inventory_ID = 102;
1 row updated.
```

```
SELECT *
FROM Inventory
WHERE Inventory_ID in(102,110);
```

INVENTORY_ID	SELLER_ID	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
102	1	102	10	In Stock	17-SEP-26
110	5	110	30	In Stock	17-SEP-26

9. Inventory Status Reports

9.1 Complete Inventory Report

```

SELECT
    i.Inventory_ID,
    s.Seller_Name,
    p.Product_Name,
    i.Stock_Quantity,
    i.Stock_Status,
    i.Last_Updated
FROM Inventory i
JOIN Seller s
ON i.Seller_ID = s.SellerID
JOIN Product p
ON i.Product_ID = p.Product_ID
ORDER BY i.Inventory_ID;

```

```

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME                                STOCK_QUANTITY
-----
STOCK_STATUS                                LAST_UPDA
-----
|      | 101
Fresh Mart
Laptop                                     50
In Stock                                16-SEP-26

```

```

INVENTORY_ID
-----
SELLER_NAME
-----
PRODUCT_NAME                                STOCK_QUANTITY
-----
STOCK_STATUS                                LAST_UPDA
-----
|      | 102
Fresh Mart
Smartphone                               10
In Stock                                17-SEP-26

```

INVENTORY_ID			

SELLER_NAME			

PRODUCT_NAME		STOCK_QUANTITY	

STOCK_STATUS		LAST_UPDA	

	103		
Daily Needs			
Tablet			30
In Stock		16-SEP-26	

INVENTORY_ID			

SELLER_NAME			

PRODUCT_NAME		STOCK_QUANTITY	

STOCK_STATUS		LAST_UPDA	

	104		
Daily Needs			
Keyboard			15
In Stock		16-SEP-26	

INVENTORY_ID			

SELLER_NAME			

PRODUCT_NAME		STOCK_QUANTITY	

STOCK_STATUS		LAST_UPDA	

	105		
Green Basket			
Mouse			0
Out of Stock		16-SEP-26	

INVENTORY_ID			

SELLER_NAME			

PRODUCT_NAME		STOCK_QUANTITY	

STOCK_STATUS		LAST_UPDA	

	106		
Green Basket			
Monitor			25
In Stock		16-SEP-26	

INVENTORY_ID			

SELLER_NAME			

PRODUCT_NAME		STOCK_QUANTITY	

STOCK_STATUS		LAST_UPDA	

107	Smart Grocery		
	Printer		40
In Stock		16-SEP-26	

INVENTORY_ID			

SELLER_NAME			

PRODUCT_NAME		STOCK_QUANTITY	

STOCK_STATUS		LAST_UPDA	

108	Smart Grocery		
	Speaker		0
Out of Stock		16-SEP-26	

INVENTORY_ID			

SELLER_NAME			

PRODUCT_NAME		STOCK_QUANTITY	

STOCK_STATUS		LAST_UPDA	

109	Nature Foods		
	Webcam		20
In Stock		16-SEP-26	

INVENTORY_ID			

SELLER_NAME			

PRODUCT_NAME		STOCK_QUANTITY	

STOCK_STATUS		LAST_UPDA	

110	Nature Foods		
	Headphones		30
In Stock		17-SEP-26	

10 rows selected.

9.2 Total Stock by Seller

```

Select
s.Seller_name,
Sum(i.Stock_Quantity)as Total_Stock
From Seller s
join Inventory i
ON s.SellerID=i.Seller_ID
Group by s.Seller_Name
Order by s.Seller_Name;

```

SELLER_NAME	

TOTAL_STOCK	

Daily Needs	
	45
Fresh Mart	
	60
Green Basket	
	25
SELLER_NAME	

TOTAL_STOCK	

Nature Foods	
	50
Smart Grocery	
	40

9.3 Product with Zero Stock

```

SELECT
    p.Product_ID,
    p.Product_Name,
    s.Seller_Name,
    i.Stock_Quantity,
    i.Stock_Status
FROM Product p
JOIN Inventory i
ON p.Product_ID = i.Product_ID
JOIN Seller s
ON i.Seller_ID = s.SellerID
WHERE i.Stock_Quantity = 0;

```

```

PRODUCT_ID PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----
    105 Mouse
Green Basket
    0 Out of Stock
    108 Speaker
Smart Grocery
    0 Out of Stock
PRODUCT_ID PRODUCT_NAME
-----
SELLER_NAME
-----
STOCK_QUANTITY STOCK_STATUS
-----

```

9.4 Available and Unavailable Product Count

```
Select Stock_Status,Count(*)as Total_Product
From Inventory
Group By Stock_Status;
```

STOCK_STATUS	TOTAL_PRODUCT
In Stock	8
Out of Stock	2

10. Conclusion

The Seller and Inventory tables were successfully created and connected with the existing Product table using primary and foreign keys. Seller and product information can be maintained along with stock quantities and stock status. Available and unavailable products can be tracked using SQL queries. Inventory reports such as complete inventory, seller-wise stock, availability, and low-stock reports can also be generated.